

Placeholder

Abstract

Strong gravitational lensing of gravitational waves is an expected consequence of the propagation of compact binary signals through an inhomogeneous universe, and a small but non-zero fraction of events detected by ground based observatories should be multiply imaged by intervening galaxies or clusters [1][2][3][4]. Recent forecasts indicate that the first confident strongly lensed gravitational wave detection could occur during the fifth LIGO-Virgo-KAGRA observing run (O5)[4], while searches incorporating O1-O4a events forecast a substantial probability of reaching at least 3σ significance in O5[5]. At the same time, simulations tailored to the wide field capabilities of the Euclid telescope demonstrate that high resolution imaging can successfully identify and localize the strongly lensed host galaxies of merging binary black holes, significantly narrowing down the gravitational wave sky localization area [6]. Motivated by these results, this work develops a Bayesian framework in which a catalogued lens with resolved strong lensing information is used to increase the confidence of a candidate lensed GW pair. We derive the full resolved strong lensing evidence, reconstruct the binary black hole population for prior odds, gravitational wave only bayes factor and joint analysis of the evidence, producing a joint bayes factor. The remaining hypotheses describe a catalogued lens without resolved images, an uncatalogued lens and two independent unlensed binary black holes. The central forecast to be tested is that targeted (EM) follow-up can, for favourable resolved strong lensing systems, sharpen the lens prediction sufficiently to raise the strong lensing detection confidence toward 5σ .

Keywords: gravitational waves; gravitational lensing: strong; Bayesian inference; electromagnetic catalogues; binary black holes; multi-messenger astronomy.

3σ to 5σ is a jump of 4500 more confident, which relates to the ratio of Bayes' factors for GW to GW+EM being 4500 but not sure if this is the same for the posterior odds as this would then assume the prior odds to be the same for GW only and joint, which we dont know if they are

1 Introduction

The field of gravitational waves (GWs) astronomy has rapidly transitioned from the first detection just about a decade ago to a large population of hundreds of detected compact-binary mergers. So far, the LIGO-Virgo-KAGRA (LVK) collaboration has reported nearly 400 confirmed GW signals from compact binary coalescences, providing an increasingly large sample with which we can use to search for rare astrophysical phenomena.

One of such phenomena is strong gravitational lensing of GWs, which is expected to occur when a GW signal passes through the gravitational potential of an intervening galaxy or cluster. Despite the growing number of detected GW events, no confident detection of strong lensing has yet been reported. Current forecasts suggest that strong lensed GW detections might be detected in the next observing run O5 [5].

Gravitational waves propagate along null geodesics and are therefore subject to gravitational lensing in the same geometric-optics sense as electromagnetic radiation [1][7]. For galaxy-scale lenses, a single compact-binary coalescence can produce multiple detectable images separated by minutes to months, depending on the lens potential and source geometry. To leading order, the images share the same intrinsic binary parameters while differing in arrival time, amplitude through the lensing magnification and an image-type-dependent phase contribution [2][8][9][10]. This repeated signal structure motivates searches for pairs or multiples of GW events that are mutually consistent under a common source hypothesis.

The central statistical difficulty is that an unlensed population can also produce event pairs with overlapping posterior distributions. A robust lensing claim must therefore compare a common-source model against a model in which the detections arise from independent binaries. Bayesian model selection provides a natural framework because it combines likelihood agreement, prior information, population expectations and parameter-space volume into a single evidence ratio [11][12][13][14]. Searches of LIGO-Virgo-KAGRA data have so far found no compelling population level evidence for strong lensing, reinforcing the need for careful prior odds and follow up tests [15][16][17].

A major source of additional information is the electromagnetic foreground. A galaxy or cluster capable of producing strong lensing may be visible in imaging or spectroscopic catalogues even when the BBH source itself has no detectable EM counterpart. The available EM information can range from simple measurements of the foreground galaxy to a resolved strong-lensing system containing arcs, multiple images or an Einstein ring from an unrelated background source. These two types of information constrain different parts of the lens model. The foreground galaxy light constrains the luminous structure and population properties of the deflector, while resolved lensed source light constrains the mapping produced by the lens mass distribution.

For this reason, the parameterisation adopted here separates the EM lens model into source light parameters, lens light parameters and lens mass parameters. This is preferable to grouping all EM quantities into a single generic lens vector because

it makes clear which parameter blocks are directly measured and which are only inferred indirectly under each hypothesis.

2 Methodology

2.1 Simulated GW events and Bayesian parameter estimation

(Consult Ankur)

2.2 Population reconstruction

The population distribution of binary black hole (BBH) parameters is crucial for determining whether two similar gravitational wave (GW) signals come from the same source, or belong to two pairs of BBH with very similar properties.

Intuitively, two GW signals are more likely to be coincidentally similar rather than being lensed if the inferred BBH parameters lie in a densely populated region of the parameter space. On the other, similar signals indicating BBH properties that are rarer in the population are more likely to be lensed.

2.2.1 Reconstruction of observed population

Population reconstruction on a simulated distribution of BBHs is performed using FIGARO, which uses a non-parametric approach to model probability densities using a Dirichlet Process Gaussian Mixture Model (DPGMM). The population density is assumed to be a sum of Gaussian components, but the number of components is determined by the data instead of being fixed in advance. As each GW event is represented by a posterior distribution from parameter estimation of a GW signal, which can be obtained from posterior samples through DPGMM, we implement the hierarchical DPGMM (HDPGMM) in FIGARO for population reconstruction (see Fig. 4, left panel, for the reconstructed population).

2.2.2 Correction for selection effects

It is important to note that the reconstructed population does not reflect the intrinsic population, as the reconstruction is based on posterior samples of detected events. In order to eliminate selection effects and create an astrophysical population, we perform rejection sampling during generation of samples from the reconstructed observational population, with acceptance probability approximated to be a function of chirp mass. The astrophysical BBH merger rate density is

$$\frac{R_{\text{GW}}(z)}{\text{Gpc}^{-3}\text{yr}^{-1}} = \frac{a_1 e^{a_2 z}}{e^{a_3 z} + a_4},$$

where $a_1 = 6600$, $a_2 = 1.6$, $a_3 = 2.1$, and $a_4 = 30$ for population-I/II stars[2].

The number of detectable events for a given chirp mass is given by

$$N_{\text{detected}}(M_c) = \int_0^{z_{\text{horizon}}(M_c)} \frac{dV_c}{dz} \frac{R(z)}{1+z} dz,$$

with $\frac{dV_c}{dz}$ being the comoving volume.
In importance sampling, the importance ratio, or the weight, is given by

$$w(x) \propto \frac{p(x)}{q(x)}, \quad x_i \sim q(x),$$

where $p(x)$ and $q(x)$ are the target population and biased population respectively.
In our case, it can be written as

$$w(M_c) \propto \frac{P_{\text{intrinsic}}(M_c)}{P_{\text{FIGARO}}(M_c)}, \quad M_c \sim P_{\text{FIGARO}}(M_c),$$

which is equivalent to

$$w(M_c) \propto \frac{P(M_c)}{P(M_c|\text{detected})}.$$

Applying Bayes theorem,

$$P(M_c|\text{detected}) = \frac{P(\text{detected}|M_c)P(M_c)}{P(\text{detected})},$$

where $P(\text{detected})$, the probability of detection, is constant, and

$$P(\text{detected}|M_c) = \frac{N_{\text{detected}}(M_c)}{N_{\text{all}}}$$

Hence, the reweighting factor can be written as

$$w(M_c) \propto \frac{1}{N_{\text{detected}}(M_c)},$$

which is then converted to acceptance probability $a(M_c)$ afterwards for rejection sampling:

$$a(M_c) = \frac{w(M_c)}{w_{\text{max}}}.$$

Rejection sampling is then performed during draws from the observed population to ensure that the new catalogue forms an astrophysical population, which is then used to compute the prior odds and Bayes factor. The astrophysical population is generally favoring lower masses due to the selection, as well as high redshifts, a trend that is critical to capture when imposing strong lensing on this population (see Fig. 4, right panel, for the reconstructed population).

2.2.3 Reconstruction of the strongly lensed population

To reconstruct the strongly lensed population, we use the LVK (LIGO-Virgo-KAGRA) Event (compact-binary mergers) Rate calculator and simulator (LeR)[18]. Samples are drawn from the reweighted unlensed population, and each of them is lensed by a foreground galaxy chosen by LeR. The lens equation is solved to generate

lensing parameters, which are saved as a dictionary. We then use the detectable events to construct the lensed population using FIGARO.

The number of lensed images depends on the source position relative to the caustics, which are boundaries in the source plane separating regions with different image multiplicities. Sources located between the outer and inner caustics generally produce two images, while those inside the inner diamond-shaped caustic produce four images. For sufficiently elliptical lenses, the diamond caustic may extend beyond the outer caustic, allowing three-image configurations. To allow further investigation and comparison between different cases, we reconstruct the population for 2-image, 3-image, and 4-image cases separately.

2.3 Prior odds estimation

The prior odds is defined to be the ratio of the probability that a given set of events are all strongly lensed images of the same source to that of the converse, rather than the ratio of the probability that a given individual event is lensed to the probability that it is not[18]. As we have corrected for selection effects in the population reconstruction, the prior odds can be calculated using

$$P_U^L = \frac{P(H_L)}{P(H_U)} \approx P(H_L) \approx N_{\text{img}}! \frac{\langle N_L \rangle}{\langle N \rangle^{N_{\text{img}}}},$$

where N_{img} is the number of images, $\langle N_L \rangle = R_L T$, and $\langle N \rangle = (R_U + N_{\text{img}} R_L) T$ [18]. For each number of images, we weigh the rate by the probability of that number of images occurring, which is determined by the lensing cross-section. The prior odds for each case are then calculated as follows:

2.4 Bayes factor computation

PO2.0

3 Bayesian hypothesis space

Consider two GW detections, $d_{GW} = d_{GW,1}, d_{GW,2}$, together with EM observations dEM drawn from a catalogue of potential foreground lenses. We divide the lensing interpretation into three mutually exclusive observational cases and compare their union with an unlensed alternative.

$$H_L = H_1 \cup H_2 \cup H_3 \tag{1}$$

with H_1 , H_2 and H_3 mutually exclusive within the assumed search space. The posterior probability of the total lensed hypothesis is therefore

$$p(H_L|d) = p(H_1|d) + p(H_2|d) + p(H_3|d) \tag{2}$$

Table 1 Mutually exclusive hypotheses used in the joint GW-EM model comparison

Hypothesis	GW interpretation	EM/catalogue information
H_1	Two detections are lensed images of the same BBH.	The responsible lens is catalogues and resolved strong-lensing EM information constrains a detailed lens model.
H_2	Two detections are lensed images of the same BBH.	The responsible lens is catalogues, but no resolved strong-lensing image configuration is available; EM data constrain galaxy properties.
H_3	Two detections are lensed images of the same BBH.	The responsible foreground lens is absent from the catalogue.
H_U	The detections arise from two independent unlensed BBHs.	Any galaxy in the search region is treated as an unrelated field object, conditional on the search selection.

where $d = d_{GW}, d_{EM}$. For each model H_i , Bayes' theorem gives

$$p(H_i|d) = Z_i \frac{p(H_i)}{p(d)}, Z_i \equiv p(d|H_i). \quad (3)$$

The posterior odds of lensing relative to the unlensed hypothesis are consequently

$$\frac{p(H_L|d)}{p(H_U|d)} = \frac{[Z_1 p(H_1) + Z_2 p(H_2) + Z_3 p(H_3)]}{Z_U p(H_U)} \quad (4)$$

Eq. (4) makes the role of the hypothesis prior explicit. The Bayes factor quantifies how the data update the relative support for the models, while the prior odds encode the expected rarity of strong lensing and the probability that a responsible lens belongs to each catalogue class. This distinction is important because prior odds and search selection can change substantially as the GW catalogue grows [11][13][17].

4 Parameterisation and Conditional Structure

4.1 Binary Black Hole Source Parameters

Let θ_{BBH} denote the physical parameters of the compact binary source. A representative set is

$$\theta_{BBH} = \{m_1, m_2, \chi_1, \chi_2, z_s, \Omega_s, \iota, \phi_{pol}, \dots\}. \quad (5)$$

Under any lensed hypothesis, the intrinsic source parameters are shared by the images, whereas under H_U the two detections have independent parameter sets $\theta_{BBH,1}$ and $\theta_{BBH,2}$. Apparent luminosity distances are image dependent because magnification rescales the observed strain amplitude.

4.2 Source light parameters

The source light block describes the resolved electromagnetic background source whose images or arcs are used to constrain the foreground lens. It does not represent an electromagnetic counterpart to the BBH. A generic source light parameter vector may be written

$$\theta_{SL} = \{\Omega_{s,EM}, F_s, R_{eff,s}, n_s, q_s, \phi_s, colour/SED_s, \dots\} \quad (6)$$

Here $\Omega_{s,EM}$ is the unlensed source-plane centroid, F_s is the source flux or surface-brightness normalisation, $R_{eff,s}$ is an effective radius, n_s is a Sersic indec, q_s is the projected axis ratio and ψ_s is the source orientation. A pixelised or basis function source reconstruction may be used instead, in that case θ_{SL} represents the corresponding source light coefficients and regularisation hyperparameters.

The source light is required only when the resolved strong lensing EM data are modelled (under H_1). It is absent from H_2 because no resolved lensed source is available, and absent from H_3 because no foreground lens candidate is identified.

4.3 Lens light parameters

The lens light block describes the luminous distribution of the foreground deflector and the observables that determine whether it is present in the catalogue. A representative parameter vector is

$$\theta_{LL} = \{\Omega_l, F_l, R_{eff,l}, n_l, q_l, \psi_l, colour/SED_l, z_l, \dots\} \quad (7)$$

These parameters describe the centroid, brightness, size, Sersic profile, axis ratio, position angle, spectral energy distribution information and redshift of the foreground galaxy. Depending on the data set, the parameters may also contain velocity dispersion measurements or population level nuisance quantities associated with photometric redshift and stellar population inference.

Under H_1 and H_2 , θ_{LL} is constrained directly by the catalogue or imaging data. Under H_3 , θ_{LL} is not observed but can be retained as a latent population variable because catalogue completeness depends primarily on the brightness, size, redshift and surface brightness properties of the foreground galaxy.

4.4 Lens mass parameters

The lens mass block describes the gravitational potential responsible for producing the multiple GW images. A model dependent vector may be written

$$\theta_{LM} = \{\theta_{E\text{or}\sigma_v}, \gamma', q_m, \phi_m, \Omega_m, \gamma_{ext}, \phi_{ext}, \kappa_{ext}, \dots\} \quad (8)$$

Here $\theta_{E\text{or}\sigma_v}$ sets the mass scale, γ' is a density slope parameter where relevant, q_m and ϕ_m describe a projected mass ellipticity and orientation, Ω_m is the mass centroid, and γ_{ext}, ϕ_{ext} and κ_{ext} describe line of sight or environmental perturbations.

The precise contents depend on whether the adopted model is, for example, a singular isothermal ellipsoid, an elliptical power law mass distribution or a more flexible composite model.

The lens mass parameters are the central bridge between GW and EM analyses. The GW pair constrains combination of θ_{LM} through the image relations, while the EM data constrain θ_{LM} either directly through resolved strong lensing morphology under H_1 or indirectly through its relation to θ_{LL} under H_2 .

4.5 Relation between the parameter blocks

The lens model parameter space is therefore organised as

$$\Theta_{lensmodel} = \{\theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \quad (9)$$

with the most important caveat that the available blocks depend on the hypothesis. The lens light and lens mass blocks need not be statistically independent. A physically motivated population prior may instead take the form

$$\pi(\theta_{LL}, \theta_{LM}) = \pi(\theta_{LM}|\theta_{LL})\pi(\theta_{LL}) \quad (10)$$

or the reverse conditioning, depending on how stellar light, velocity dispersion, halo mass and mass to light relations are modelled. Treating this correlation explicitly is especially important under H_2 , where the lens light supplies the principle EM route to constraining lens mass.

4.6 Image Level Lensing Observables

The GW pair constrains a lower dimensional set of relations between the images,

$$\psi_{GW} = \{\Delta t, \mu_r, \Delta n, \Omega, \dots\} \quad (11)$$

where Δt is the relative arrive time delay, μ_r is a relative magnification ratio, Δn denotes the relative Morse index, and Ω represents common sky localisation. These quantities are generated by the lens equation and Fermat potential rather than being independent parameters,

$$\psi = f(\theta_{LM}, z_s, \beta; cosmology). \quad (12)$$

4.7 Electromagnetic Nuisance Parameters

EM observations require nuisance parameters ν_{EM} describing measurement and modelling uncertainties. Depending on the catalogue and imaging data, ν_{EM} may include the point spread function, photometric calibration, spectral energy distribution parameters, stellar mass to light ratio, background subtraction, photometric redshift calibration and other instrument or population level terms. These parameter should be

magnalised rather than fixed when their uncertainty is non negligible, beause overly narrow EM constraints can artificially increase the GW-EM consistency factor.

5 Unlensed Hypothesis

Under H_U the detections correspond to two independent BBH mergers. The GW evidence is

$$Z_{GW,U} = \int p(d_{GW,1}|\theta_{BBH,1}, H_U) p(d_{GW,2}|\theta_{BBH,2}, H_U) \pi_U(\theta_{BBH,1}, \theta_{BBH,2}) d\theta_{BBH,1} d\theta_{BBH,2} \quad (13)$$

The EM observations must not be allowed to depend pn whether the GW pair happens to be unlensed. We therefore introduce an EM field model, $H_{EM,field}$, that describes unrelated foreground galaxies under the same catalogue/search selection. When the GW and EM data are conditionally independent under H_U ,

$$H_U \equiv H_{GW,U} \cap H_{EM,field} \quad (14)$$

$$Z_U = Z_{GW,U} Z_{EM,field}, Z_{EM,field} = p(d_{EM}|H_{EM,field}) \quad (15)$$

This term is essential whenever the lensed hypotheses recieve evidence from the existence or properties of a catalogue galaxy. Omitting the corresponding field galaxy probability from H_U would systematically favour catalogues lens hypotheses simply because a galaxy was searched for and found.

6 H_1 : catalogued lens with resolved strong lensing information

6.1 Resolved EM data and parameter set

Under H_1 , the two GW detections are strongly lensed images of the same BBH and the responsible foreground lens is present in the EM catalogue with resolved strong lensing information. (not sure about this but had a thought) The resolved EM source need not be the BBH counterpart; it provides an independednt probe of the same foreground gravitational potential. The EM data may therefore be separated schematically into measurements of the lens galaxy and measurements of the resolved strong lensing configuration.

$$d_{EM,c} = \{d_{LL,c}, d_{SL,c}\} \quad (16)$$

where $d_{LL,c}$ denotes measurements of the foreground lens light, while $d_{SL,c}$ denotes the resolved image positions, arc, an Einstein ring or pixel level surface brightness data.

The parameter vector is

$$\Theta_1 = \{\theta_{BBH}, \beta_{GW}, \theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \quad (17)$$

The central coupling is θ_{LM} . The GW likelihood depends on θ_{LM} through the relative image properties, while the resolved EM likelihood constrains θ_{LM} through the mapping between θ_{SL} and the observed lensed surface brightness. The lens light parameters θ_{LL} model the foreground luminous component and can also inform physically motivated priors on θ_{LM} .

6.2 Full joint evidence and prior

$$p(d_{GW}, d_{EM,c} | \Theta_1, H_1) = p(d_{GW} | \theta_{BBH}, \theta_{LM}, \beta_{GW}, H_1) p(d_{EM,c} | \theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1). \quad (18)$$

The full evidence for candidate c is

$$Z_{1,c} = \int p(d_{GW} | \theta_{BBH}, \theta_{LM}, \beta_{GW}, H_1) p(d_{EM,c} | \theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1) \pi_1(\Theta_1) d\Theta_1. \quad (19)$$

A useful schematic prior factorisation is

$$\pi_1(\Theta_1) = \pi_{BBH} \pi_{\beta_{GW}} \pi_{SL} \pi(\theta_{LL}, \theta_{LM}) \pi_{\nu_{EM}} S_1. \quad (20)$$

The joint prior $\pi(\theta_{LL}, \theta_{LM})$ allows correlations between galaxy light and mass to be retained. S_1 denotes the required GW detection, catalogue and resolved lens selection conditioning. The evidence integrated over both the scientifically interesting lens mass parameters and the source light, lens light and nuisance parameters required by the EM reconstruction.

6.3 GW observables and the EM predictive distribution

The resolved EM data determine a posterior for the lens mass after marginalising over source light, lens light and EM nuisance parameters:

$$p(\theta_{LM} | d_{EM,c}, H_1) \propto \int p(d_{EM,c} | \theta_{SL}, \theta_{LL}, \theta_{LM}, \nu_{EM}, H_1) \pi_1 d\theta_{SL} d\theta_{LL} d\nu_{EM} \quad (21)$$

Propagating this mass posterior through the lens equation gives the predictive distribution

$$p(\psi | d_{EM,c}, H_1) = \int p(\psi | \theta_{LM}, z_s, \beta_{GW}, H_1) p(\theta_{LM}, z_s, \beta_{GW} | d_{EM,c}, H_1) d\theta_{LM} dz_s d\beta_{GW} \quad (22)$$

The corresponding prior corrected consistency statistic is

$$C_1 = \int \frac{p(\psi | d_{GW}, H_1) p(\psi | d_{EM,c}, H_1)}{\pi(\psi | H_1)} d\psi \quad (23)$$

A large C_1 indicates that the resolved EM reconstruction predicts time delays, magnification relations and image properties compatible with the GW pair. A small value

indicates tension. When the full evidence in Eq. (19) is evaluated directly, this consistency is already encoded through θ_{LM} ; C_1 is therefore most safely interpreted as a reduced space diagnostic unless an explicit non overlapping factorisation is derived.

6.4 Multiple resolved lens candidates

If more than one resolved strong lensing system is compatible with the GW localisation, the lens identity should be marginalised rather than fixed in advance:

$$Z_1 = \sum_{c=1}^{N_{cat}} p(c|H_1) Z_{1,c} \quad (24)$$

7 Reduced GW-EM consistency under H_1

The reduced consistency statistic isolates the direct comparison between the GW inferred image relations and the prediction from the resolved EM lens reconstruction. Its prior denominator removes the common prior contribution that would otherwise be counted twice when multiplying posteriors.

In the present parameterisation, the key reason H_1 can be highly constraining is that the resolved source light morphology θ_{SL} is lensed by the same θ_{LM} that generates the GW image relations. Marginalising over θ_{SL} and θ_{LL} can still leave a comparatively narrow posterior on θ_{LM} , provided the strong lensing configuration is informative.

8 Computational evaluation of H_1

The integrals can be manipulated and approximated using laplace to get a form more suitable for computational analysis. Using the joint maximum and laplace approximation the full evidence for H_1 can be written, in log form as,

$$\ln(Z_{1,c}) \approx \ell_1(\hat{\Theta}_1) + \left(\frac{k_1}{2}\right) \ln(2\pi) - \left(\frac{1}{2}\right) \ln(|A_1|) \quad (25)$$

Eq. (25) separates the evidence into a best fit contribution, $\ell_1(\hat{\Theta}_1)$, and a local posterior volume contribution determined by k_1 and $|A_1|$. The resolved EM data can alter both the peak term and the local posterior volume. In the source light/lens light/lens mass parameterisation, the full Hessian contains cross block curvature. In particular, degeneracies between source light and lens mass, and between lens light and lens mass, should be retained rather than ignored by approximating the blocks independently.

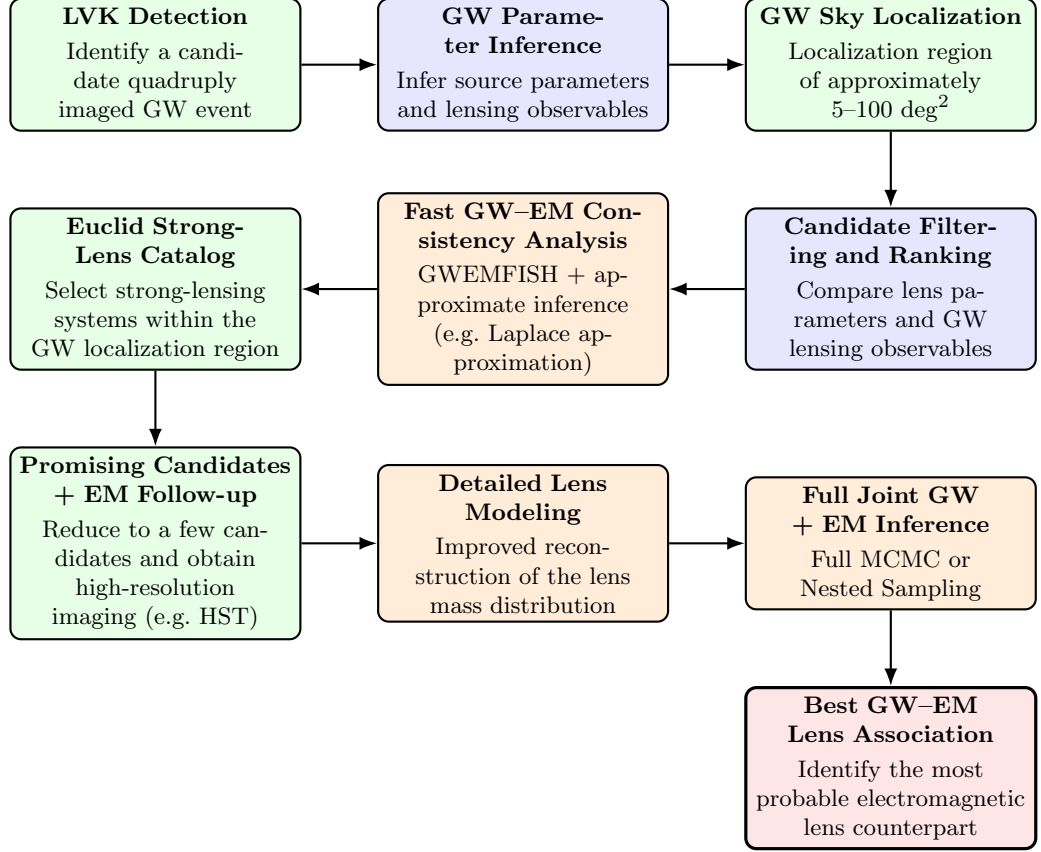


Figure 1 Conceptual workflow for identifying the electromagnetic lens counterpart of a quadruply imaged gravitational-wave event.

9 Joint Bayes factor and posterior odds

9.1 GWEMFISH and Laplace’s Approximation: A physicist’s life (and time) saver

Lens reconstruction will usually take very long (from hours to days) if it was ran on a traditional Markov Chain Monte Carlo program. To overcome this issue and ensure that the lens reconstruction can be computed quickly, we use GWEMFish, which utilizes Laplace approximation, to ensure that each computation is turbo-fast. -more details or any references to other sections on how this works

Using Laplace’s approximation, we can approximate the evidence for each lens candidate in the GW localization region. This allows us to quickly compute the Bayes factor for each candidate and rank them based on their likelihood of being the correct lens. With this, we can much more easily narrow down the candidates and focus on the most promising ones for further analysis and follow-up observations, such as

performing a full MCMC or Nested Sampling and follow up using a telescope with higher resolution such as the HST.

9.2 Lens searches and the localization problem

It is known that GWs poor localization compared to other EM counterparts (normally around $5 - 25 \text{deg}^2$). In a region of sky of that size, it is predicted by (referece) that the Euclid telescope could detect up to around 600 lensing galaxies. This means that we can't immediately know which galaxy the GW signal originates. This is analogous to the birthday problem where we would need to perform searches with the lensing candidates in the localized region of sky to match the GW signal to the correct lens. Previous studies (refereces) have concluded that the correct lens would equate to being able to localize the GW correctly.

9.3 From an omniscient point of view: the joint Bayes factor

10 Result

10.1 Reconstructed Population

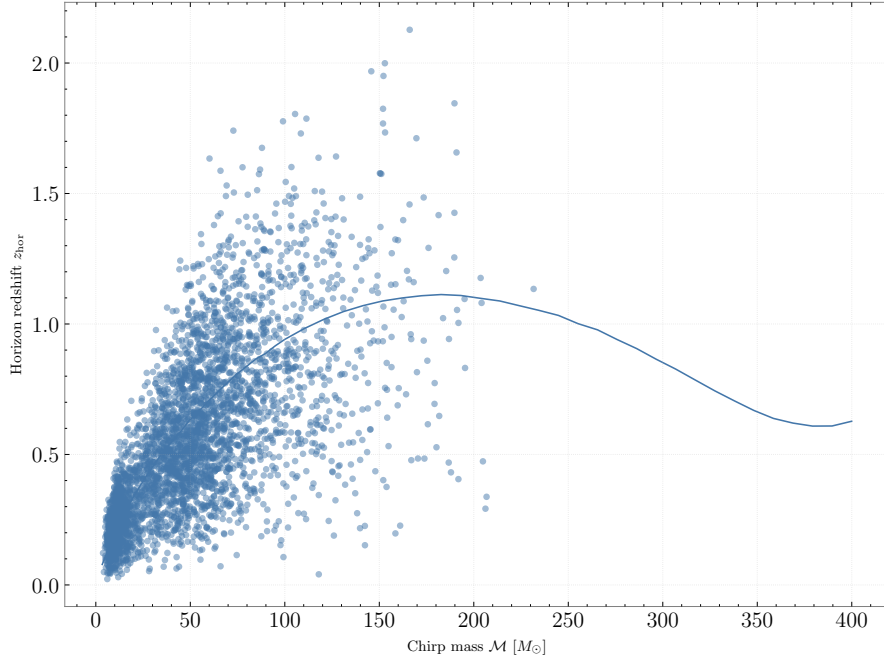


Figure 2

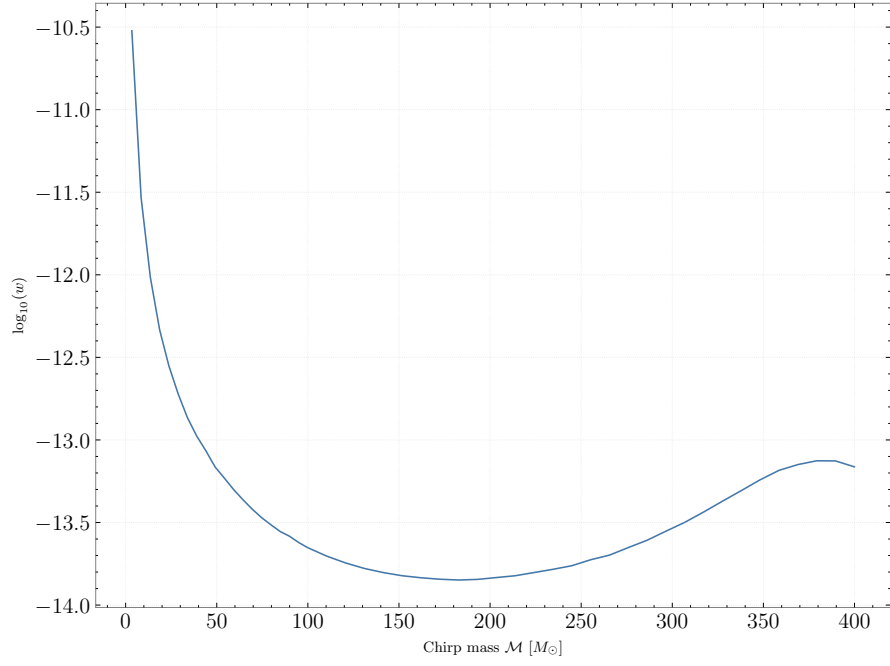


Figure 3

10.2 Prior Odds

rates, rate ratio, prior odds.

Results:

2-image lensed rate: 0.21 detectable events per year

2-image rate ratio: 0.00077

2-image prior odds: 5.6e-06

3-image lensed rate: 0.04 detectable events per year

3-image rate ratio: 0.00016

3-image prior odds: 1.3e-08

4-image lensed rate: 0.16 detectable events per year

4-image rate ratio: 0.00057

4-image prior odds: 6.6e-10

10.3 Bayes Factor and Comparison

11 Conclusion

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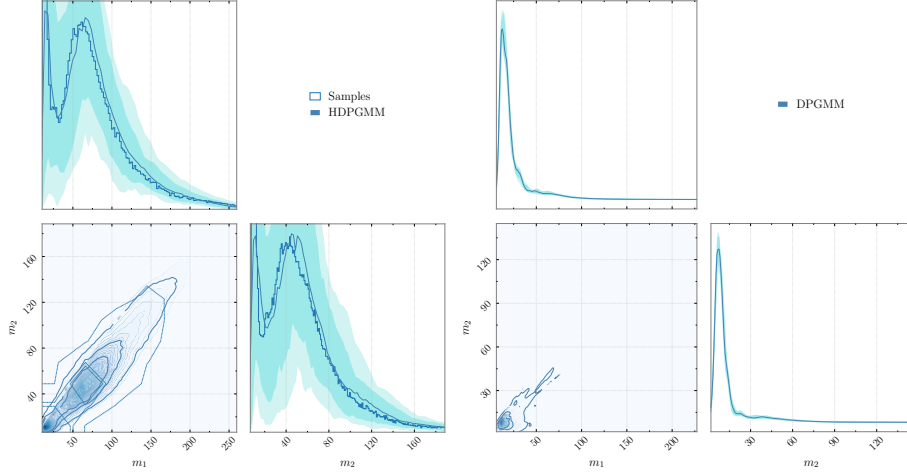


Figure 4 Reconstructed observed (left panel) and astrophysical (right panel) binary black hole population in component masses m_1 and m_2 using the hierarchical DPGMM implementation in FIGARO inferred from simulated population of gravitational-wave signals mimicking observing run O4. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The observed population is biased towards higher masses due to selection effects, while the astrophysical population is corrected for these effects and represents the true underlying distribution of binary black hole mergers, extending to higher redshifts and lower masses that are relevant to strong gravitational-wave lensing, which occurs for binaries at high redshift.

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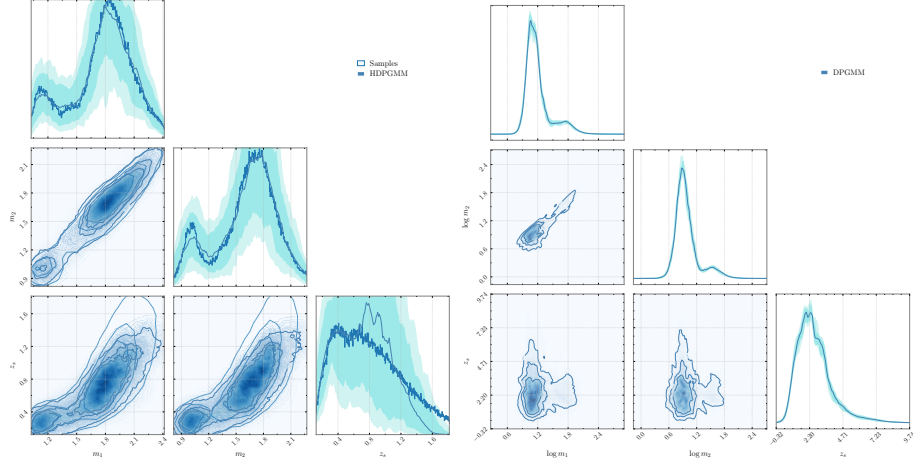


Figure 5 Reconstructed observed (left panel) and astrophysical (right panel) binary black hole population in $\log_{10}$ of component masses m_1 and m_2 using the hierarchical DPGMM implementation in FIGARO inferred from simulated population of gravitational-wave signals mimicking observing run O4. The diagonal panels show the one-dimensional marginal distributions, while the off-diagonal panels show the corresponding two-dimensional joint distributions. The observed population is biased towards higher masses due to selection effects, while the astrophysical population is corrected for these effects and represents the true underlying distribution of binary black hole mergers, extending to higher redshifts and lower masses that are relevant to strong gravitational-wave lensing, which occurs for binaries at high redshift.

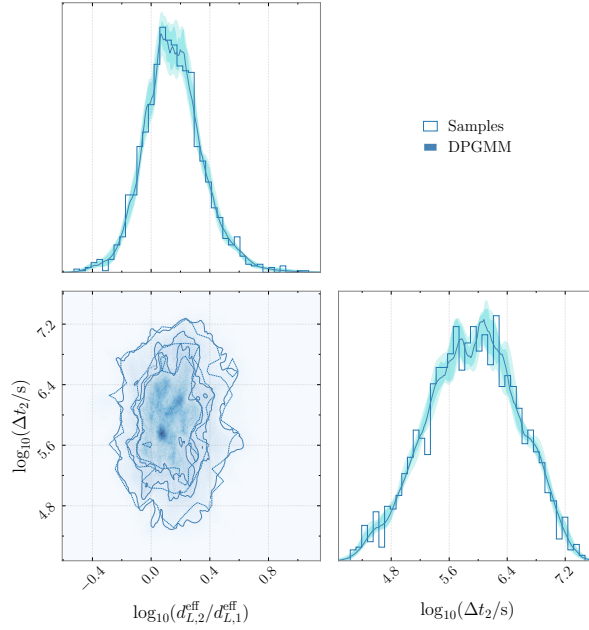


Figure 6

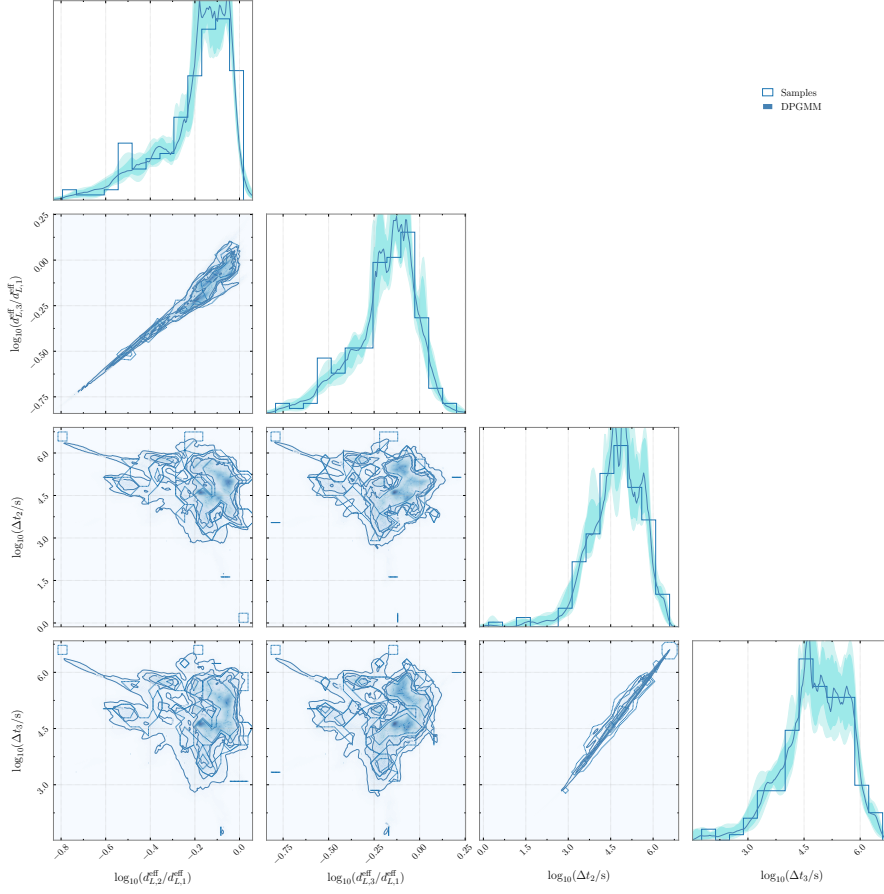


Figure 7

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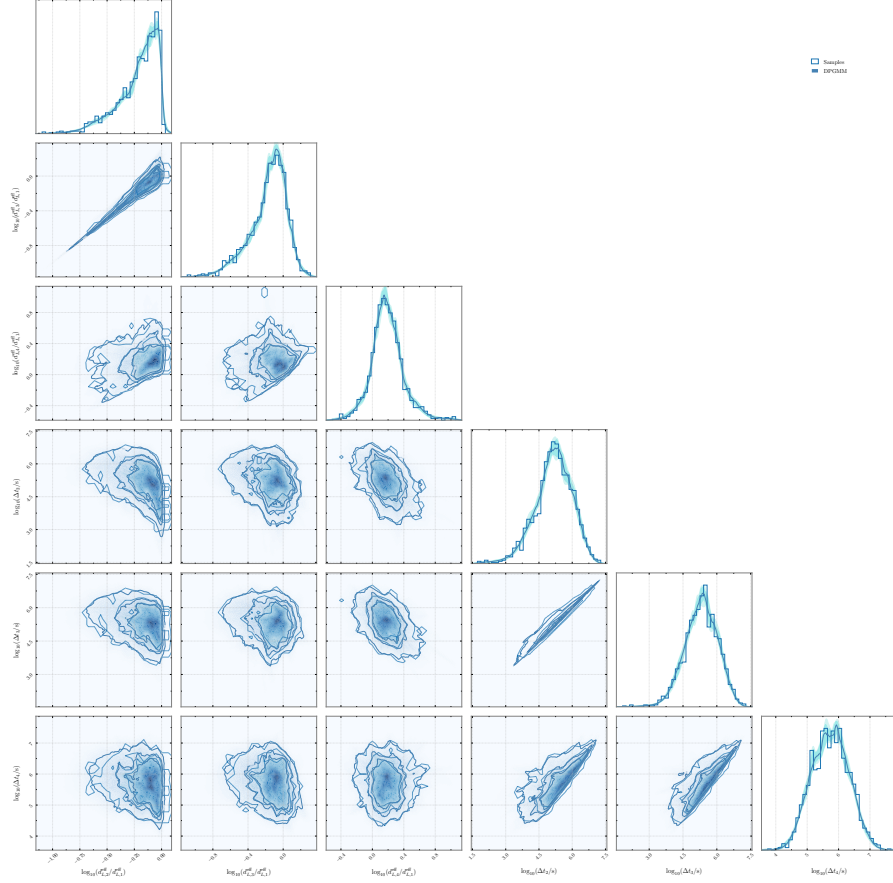


Figure 8

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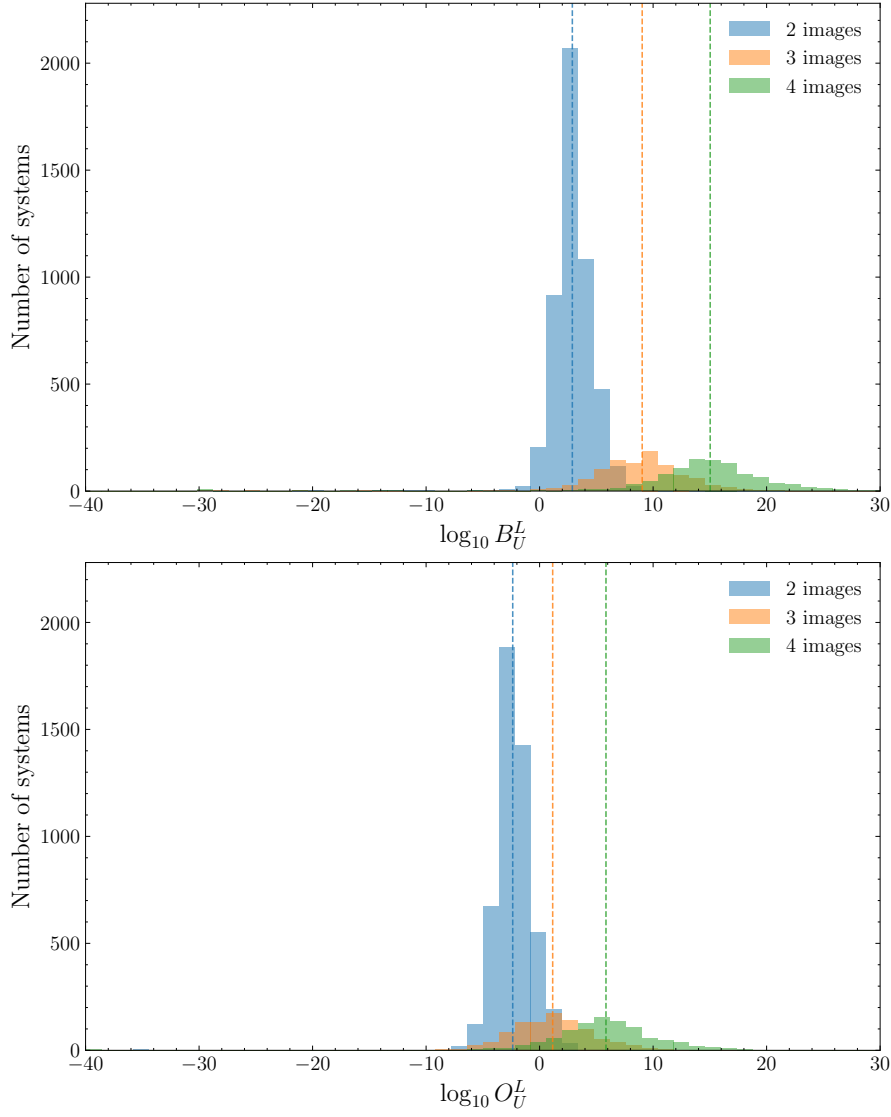


Figure 9

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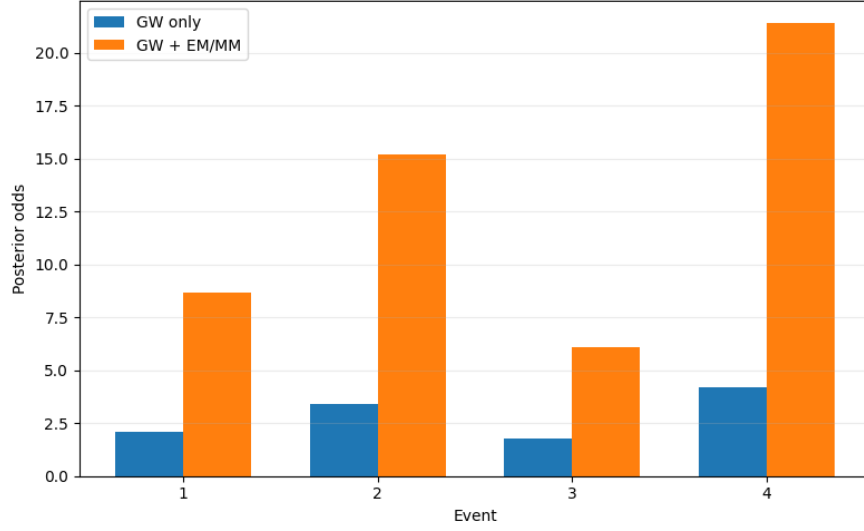


Figure 10

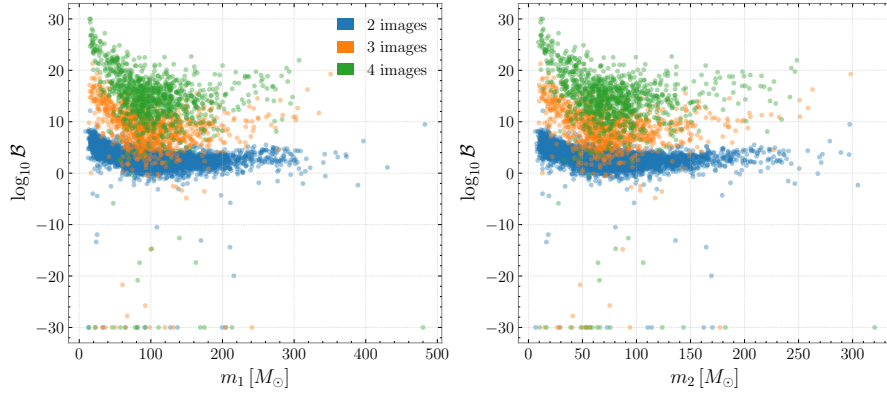


Figure 11

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Appendix A H_2 : catalogued lens without resolved strong lensing EM images

Available parameter blocks

H_2 retains the common source lensing interpretation but removes the resolved source light information. The foreground galaxy is catalogued and its lens light parameters are constrained by imaging, photometry and spectroscopy, but no resolved background source is available from which to reconstruct the lens mass directly.

$$\Theta_2 = \{\theta_{BBH}, \beta_{GW}, \theta_{LL}, \theta_{LM}, \nu_{EM}\} \quad (\text{A1})$$

The absence of θ_{SL} is the defining parameter space difference from H_1 . Lens mass is constrained only through the GW data and through the population or scaling relation connecting θ_{LL} to θ_{LM} .

Full evidence

$$p(d_{GW}, d_{EM,c} | \Theta_2, H_2) = p(d_{GW} | \theta_{BBH}, \theta_{LM}, \beta_{GW}, H_2) p(d_{EM,c} | \theta_{LL}, \nu_{EM}, H_2) \quad (\text{A2})$$

Because the catalogue likelihood primarily constrains θ_{LL} , the connection to θ_{LM} is carried by the joint prior $\pi(\theta_{LL}, \theta_{LM})$, rather than by a resolved lensing likelihood. The full evidence is

$$Z_{2,c} = \int p(d_{GW} | \theta_{BBH}, \theta_{LM}, \beta_{GW}, H_2) p(d_{EM,c} | \theta_{LL}, \nu_{EM}, H_2) \pi_2(\Theta_2) d\Theta_2 \quad (\text{A3})$$

with a schematic prior

$$\pi_2(\Theta_2) = \pi_{BBH} \pi_{\beta_{GW}} \pi(\theta_{LL}, \theta_{LM} | \pi_{\nu_{EM}}) S_2 \quad (\text{A4})$$

This form is preferable to treating the EM likelihood as if it measured θ_{LM} directly. Any inference from luminosity, effective radius, redshift or velocity dispersion to lens mass should enter through the appropriate conditional population model.

EM induced lens mass and image observable distributions

The catalogue information induces a lens mass posterior by marginalising over the measured lens light parameters:

$$p(\theta_{LM}|d_{EM,c}, H_2) \propto \int p(d_{EM,c}|\theta_{LL}, \nu_{EM}, H_2) \pi(\theta_{LL}, \theta_{LM}) \pi_{\nu_{EM}} d\theta_{LL} d\nu_{EM} \quad (\text{A5})$$

This distribution can then be propagated to the GW image observables:

$$p(\psi|d_{EM,c}, H_2) = \int p(\psi|\theta_{LM}, z_s, \beta_{GW}, H_2) p(\theta_{LM}, z_s, \beta_{GW}|d_{EM,c}, H_2) d\theta_{LM} dz_s d\beta_{GW} \quad (\text{A6})$$

Relative to H_1 , this distribution is generally broader because θ_{LM} is inferred indirectly from lens light rather than from a resolved source light configuration.

Reduced consistency statistic

$$C_2 = \int \frac{p(\psi|d_{GW}, H_2) p(\psi|d_{EM,c}, H_2)}{\pi(\psi|H_2)} d\psi \quad (\text{A7})$$

As under H_1 , C_2 is best regarded as a diagnostic when the full joint evidence is already evaluated, because the common lens mass information is then already included in Eq. (A3).

Laplace approximation

$$\ell_2(\Theta_2) = \ln(p(d_{GW}|\Theta_2, H_2)) + \ln(p(d_{EM,c}|\Theta_2, H_2)) + \ln(\pi_2(\Theta_2)) \quad (\text{A8})$$

$$\hat{\Theta}_2 = \arg \max_{\Theta_2} \ell_2(\Theta_2) \quad (\text{A9})$$

$$A_2 = -\nabla \nabla \ell_2(\Theta_2)|_{\hat{\Theta}_2} \quad (\text{A10})$$

$$\ln(Z_{2,c}) \simeq \ell_2(\hat{\Theta}_2) + \left(\frac{k_2}{2}\right) \ln(2\pi) - \left(\frac{1}{2}\right) \ln(|A_2|) \quad (\text{A11})$$

The curvature matrix A_2 contains cross information between θ_{LL} and θ_{LM} through their population relation. Because no θ_{SL} block is present, the EM data does not provide the strong image geometry constraints available under H_1 . The resulting posterior in the lens mass directions will therefore normally be broader and more prior dependent.

Restriction of H_2

H_2 is restricted to systems in which the responsible lens is present in the catalogue but no resolved strong lensing EM configuration is used. If a resolved lensed source is available, the system belongs to H_1 , if the responsible foreground lens is absent from the catalogue, it belongs to H_3 .

The principle physical limitation is that lens light does not uniquely determine lens mass. The relation $\pi(\theta_{LM}|\theta_{LL})$ must therefore include intrinsic scatter and relevant uncertainties. Treating the lens light measurement as a deterministic mass estimate would artificially narrow the H_2 evidence.

The source redshift and GW source plane position remain latent unless measure independently. The Laplace approximation also requires an approximately single mode, locally Gaussian posterior. Weak lens mass constraints and light to mass degeneracies can make this assumption poor.

Appendix B H_3 : uncatalogued lens

Latent lens light and lens mass population

Under H_3 , the GW pair is lensed but the responsible foreground lens is not present in the catalogue. No candidate specific light source or lens light likelihood exists. Nevertheless, retaining a latent θ_{LL} block is useful because catalogue detection depends on the luminous properties of the deflector, while the lensing signal depends on θ_{LM} .

$$\Theta_3 = \{\theta_{BBH}, \beta_{GW}, \theta_{LL}, \theta_{LM}\} \quad (B12)$$

The catalogue completeness function is written

$$C_{cat}(\theta_{LL}, \theta_{LM}) = P(\text{lens appears in catalogue} | \theta_{LL}, \theta_{LM}) \quad (B13)$$

where dependence on θ_{LM} may be retained when survey selection is correlated with mass through velocity dispersion, halo mass or environmental properties. In the simplest implementation the completeness may depend only on the observable lens light block.

Full evidence

The H_3 evidence becomes

$$Z_3 = \int p(d_{GW} | \theta_{BBH}, \theta_{LM}, \beta_{GW}, H_3) [1 - C_{cat}(\theta_{LL}, \theta_{LM})] \pi_{BBH} \pi_{\beta_{GW}} \pi(\theta_{LL}, \theta_{LM}) d\Theta_3 \quad (B14)$$

This form makes the parameter space structure explicit. The GW likelihood selects lens mass configurations capable of producing the observed image relations, the joint lens light.lens mass population prior supplies physically plausible deflectors, and the non-detection factor suppresses systems whose light properties imply that they should have appeared in the catalogue.

Log integrand and laplace approximation

$$\ell_3(\Theta_3) = \ln(p(d_{GW}|\theta_{BBH}, \theta_{LM}, \beta_{GW}, H_3)) + \ln([1 - C_{cat}(\theta_{LL}, \theta_{LM})]) + \ln(\pi_{BBH}) + \ln(\pi_{\beta_{GW}}) + \ln(\pi(\theta_{LL}, \theta_{LM})) \quad (\text{B15})$$

$$\hat{\Theta}_3 = \arg \max_{\Theta_3} \ell_3(\Theta_3) \quad (\text{B16})$$

$$A_3 = -\nabla \nabla \ell_3(\Theta_3)|_{\hat{\Theta}_3} \quad (\text{B17})$$

$$\ln(Z_3) \simeq \ell_3(\hat{\Theta}_3) + \left(\frac{k_3}{2}\right) \ln(2\pi) - \left(\frac{1}{2}\right) \ln(|A_3|) \quad (\text{B18})$$

The catalogue non-detection term can alter both the height of the dominant mode and the curvature of the posterior. A lens mass solution that fits the GW well can still be suppressed if its correlated lens light properties make non-detection improbable.

Why there is no source light consistency factor under H_3

There is no candidate specific resolved background source under H_3 , and therefore no source light block θ_{SL} and no EM reconstruction from which to construct $p(\psi|d_{EM,c}, H_3)$. The role played by C_1 and C_2 is replaced by marginalisation over the latent lens light/lens mass population and the catalogue non-detection probability.

Restriction of H_3

H_3 applies only when the responsible lens is absent from the adopted catalogue. The completeness model must therefore refer to exactly the same survey footprint, depth and object selection criteria used to define H_1 and H_2 .

The joint population model $\pi(\theta_{LL}, \theta_{LM})$ is essential. If lens light is omitted completely, the catalogue non-detection probability must be expressed indirectly as a function of mass, which can hide substantial astrophysical uncertainty. Retaining θ_{LL} as a latent variable makes the selection model more transparent.

The laplace approximation can fail when catalogue completeness changes sharply near a survey threshold, when the light to mass population relation is strongly non-Gaussian or when several lens mass configurations provide comparable fits to the GW data.

Appendix C Laplace approximation to the full H_1 evidence

Define the H_1 integrand

$$I_1(\Theta_1) = p(d_{GW}|\Theta_1, H_1)p(d_{EM,c}|\Theta_1, H_1)\pi_1(\Theta_1) \quad (\text{C19})$$

so that

$$Z_{1,c} = \int I_1(\Theta_1) d\Theta_1 \quad (\text{C20})$$

For Numerical stability, computations are performed in log space:

$$\ell_1(\Theta_1) = \ln(p(d_{GW}|\Theta_1, H_1)) + \ln(p(d_{EM,c}|\Theta_1, H_1)) + \ln \pi_1(\Theta_1) \quad (\text{C21})$$

$$Z_{1,c} = \int e^{[\ell_1(\Theta_1)]} d\Theta_1 \quad (\text{C22})$$

The maximum a posteriori point is

$$\hat{\Theta}_1 = \arg \max_{\Theta_1} \ell_1(\Theta_1) \quad (\text{C23})$$

Near $\hat{\Theta}_1$, expand the log integrand to second order. Since the gradient vanishes at an interior maximum, the local form is

$$\ell_1(\Theta_1) \approx \ell_1(\hat{\Theta}_1) - \frac{1}{2}(\Theta_1 - \hat{\Theta}_1)^T A_1 (\Theta_1 - \hat{\Theta}_1) \quad (\text{C24})$$

with negative Hessian

$$A_1 = -\nabla \nabla \ell_1(\Theta_1)|_{\hat{\Theta}_1} \quad (\text{C25})$$

Substitutes into Eq. (C22) gives a multivariate Gaussian integral:

$$Z_{1,c} \approx e^{[\ell_1(\hat{\Theta}_1)]} (2\pi)^{\frac{k_1}{2}} |A_1|^{-\frac{1}{2}} \quad (\text{C26})$$

where k_1 is the number of free continuous parameters. In log form,

$$\ln(Z_{1,c}) \approx \ell_1(\hat{\Theta}_1) + \left(\frac{k_1}{2}\right) \ln(2\pi) - \left(\frac{1}{2}\right) \ln(|A_1|) \quad (\text{C27})$$

Appendix D Gaussian approximation and Fisher matrix in the consistency statistic

Recall the reduced consistency statistic

$$C_1 = \int \frac{p(\psi|d_{GW}, H_1)p(\psi|d_{EM}, H_1)}{\pi(\psi|H_1)} d\psi \quad (\text{D28})$$

Posterior overlap statistics of this general form are widely used in Bayesian searches for strongly lensed GW event pairs [19], while later lensing frameworks perform more complete joint parameter inference and evidence calculations [20][13].

Assume that the GW and EM distributions are locally well approximated by k_ψ dimensional Gaussian densities,

$$p(\psi|d_{GW}, H_1) \simeq \mathcal{N}(\psi; \mu_{GW}, \Sigma_{GW}) \quad (\text{D29})$$

$$p(\psi|d_{EM}, H_1) \simeq \mathcal{N}(\psi; \mu_{EM}, \Sigma_{EM}) \quad (\text{D30})$$

If the prior varies slowly across the region in which both distributions have appreciable support, it may be approximated locally by a constant $\pi(\psi|H_1) \simeq \pi_0$. The consistency statistic then becomes

$$C_1 \simeq \frac{1}{\pi_0} \int \mathcal{N}(\psi; \mu_{GW}, \Sigma_{GW}) \mathcal{N}(\psi; \mu_{EM}, \Sigma_{EM}) d\psi \quad (\text{D31})$$

The product integral of two normalised multivariate Gaussian densities is

$$\int \mathcal{N}(\psi; \mu_1, \Sigma_1) \mathcal{N}(\psi; \mu_2, \Sigma_2) = \mathcal{N}(\mu_1; \mu_2, \Sigma_1 + \Sigma_2) \quad (\text{D32})$$

Therefore

$$C_1 \simeq \frac{1}{\pi_0} \mathcal{N}(\mu_{GW}; \mu_{EM}, \Sigma_{GW} + \Sigma_{EM}) \quad (\text{D33})$$

Writing the Gaussian density explitley gives

$$C_1 \simeq \frac{e^{-\frac{D^2}{2}}}{\pi_0 (2\pi)^{\frac{k_\psi}{2}} |\Sigma_{GW} + \Sigma_{EM}|^{\frac{1}{2}}} \quad (\text{D34})$$

with

$$D^2 = (\mu_{GW} - \mu_{EM})^T (\Sigma_{GW} + \Sigma_{EM})^{-1} (\mu_{GW} - \mu_{EM}) \quad (\text{D35})$$

This is the squared Mahalanobis distance [21] between the two posterior means, evaluated using their combined covariance.

Fisher information provides a complementary description of measurement precision. For a likelihood $L(d|\psi)$ the expected Fisher matrix is defined by [22]

$$F_{ij} = - \left\langle \frac{\partial^2 \ln L}{\partial \psi_i \partial \psi_j} \right\rangle \quad (\text{D36})$$

under the usual regularity assumptions. Near a well constrained maximum, the log likelihood may be expanded as

$$\ln(L(\psi)) \simeq \ln(L(\hat{\psi})) - \frac{1}{2} (\psi - \hat{\psi})^T F (\psi - \hat{\psi}) \quad (\text{D37})$$

In the asymptotic Gaussian limit, and when prior curvature is negligible,

$$\Sigma \simeq F^{-1} \quad (\text{D38})$$

This approximation is standard in GW parameter estimation, but its domain of validity must be checked carefully: finite SNR ratio, weakly measured parameters, boundaries, degeneracies and informative priors can all make the Fisher approximation inaccurate [23].

For the reduced consistency problem,

$$\Sigma_{GW} \simeq F_{GW}^{-1}, \Sigma_{EM} \simeq F_{EM}^{-1}. \quad (\text{D39})$$

The total covariance becomes

$$\Sigma_{tot} \simeq F_{GW}^{-1} + F_{EM}^{-1}. \quad (\text{D40})$$

Thus the Mahalanobis distance may be written directly in terms of the two information matrices,

$$D^2 = \Delta\mu^T (F_{GW}^{-1} + F_{EM}^{-1})^{-1} \Delta\mu \quad (\text{D41})$$

Substitution into the Gaussian overlap gives.

$$C_1 \simeq \frac{e^{-\frac{1}{2} \Delta\mu^T (F_{GW}^{-1} + F_{EM}^{-1})^{-1} \Delta\mu}}{\pi_0 (2\pi)^{\frac{k_\psi}{2}} |F_{GW}^{-1} + F_{EM}^{-1}|^{\frac{1}{2}}} \quad (\text{D42})$$

Taking the logarithm yields a numerically convenient decomposition,

$$\ln(C_1) \simeq -\ln(\pi_0) - \frac{k_\psi}{2} \ln(2\pi) - \frac{1}{2} \ln(|F_{GW}^{-1} + F_{EM}^{-1}|) - \frac{1}{2} D^2 \quad (\text{D43})$$

The last two terms have distinct interpretations:

1. $-\frac{D^2}{2}$ is the mismatch term. It penalises separation between the GW and EM predictions in units set by the combined uncertainty.
2. $-\frac{1}{2} \ln(|F_{GW}^{-1} + F_{EM}^{-1}|)$ is the precision volume term. It increases as the joint uncertainty volume becomes smaller.

This decomposition makes clear why resolved strong lensing EM information can be highly discriminating under H_1 . If the EM lens reconstruction is precise, F_{EM} has large eigenvalues and F_{EM}^{-1} is small. Agreement with the GW posterior can then produce a sharply concentrated overlap. Conversely, a precise EM prediction that is offset from the GW inference can make D^2 large and suppress C_1 exponentially.